NHS Dorset Data Scientist technical assessment — references and tools

Tools used in the analysis (confirm against your own notebooks before submission):

- Python 3.12: programming language for data preparation, analysis and modeling.

- pandas and NumPy: profiling, cleaning, feature engineering and data manipulation.

- scikit-learn: preprocessing pipelines, Logistic Regression, Random Forest, evaluation metrics and model validation.

- XGBoost: gradient-boosted classifier and hyperparameter tuning.

- matplotlib and seaborn: exploratory and evaluation visualizations.

- Jupyter Notebook / VS Code: developing, executing and documenting analysis.

- joblib: saving and reloading the fitted model pipeline.

- Git / GitHub: project version control and organization.

- pandas documentation: https://pandas.pydata.org/docs/

- scikit-learn user guide: https://scikit-learn.org/stable/user_guide.html

- XGBoost documentation: https://xgboost.readthedocs.io/

- NumPy documentation: https://numpy.org/doc/

- matplotlib documentation: https://matplotlib.org/stable/

- seaborn documentation: https://seaborn.pydata.org/

Disclosure: These are software tools and potential documentation references, not a claim that any particular tutorial or public code was copied. Before sending, remove documentation links you did not actually consult, and add any external tutorials or public code you used, explaining the specific use and attribution. The supplied synthetic dataset came from NHS Dorset.